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1 Betti Numbers and Euler Characteristic

Claim: Let

$$0 \rightarrow A_1 \xrightarrow{f_1} A_2 \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} A_n \rightarrow 0$$

be an exact sequence of finitely generated Abelian groups. Then $\sum_{i=1}^n (-1)^i \text{rank } A_i = 0$.

Proof: Note that $A_i / \ker f_i \cong \text{im } f_i \rightarrow A_i / \ker f_i \cong \ker f_{i+1} \rightarrow \text{rank } A_i - \text{rank } \ker f_i = \text{rank } \ker f_{i+1} \rightarrow \text{rank } \ker f_i + \text{rank } \ker f_{i+1} = \text{rank } A_i$.

I claim by induction that $\text{rank } \ker f_k = \sum_{i=1}^{k-1} (-1)^{i+1} \text{rank } A_{k-i}$. Indeed, $\text{rank } \ker f_1 = 0$ since f_1 is injective and

$$\begin{aligned} \text{rank } \ker f_k &= \text{rank } A_{k-1} - \text{rank } \ker f_{k-1} = \text{rank } A_{k-1} - \sum_{i=1}^{k-2} (-1)^{i+1} \text{rank } A_{k-1-i} \\ &= \text{rank } A_{k-1} + \sum_{i=1}^{k-2} (-1)^i \text{rank } A_{k-1-i} = \text{rank } A_{k-1} + \sum_{i=2}^{k-1} (-1)^{i+1} \text{rank } A_{k-i} \\ &= \sum_{i=1}^{k-1} (-1)^{i+1} \text{rank } A_{k-i}. \end{aligned}$$

In particular, $\ker f_{n+1} = 0$ (by augmenting zero groups), so

$$\sum_{i=1}^n (-1)^{i+1} \text{rank } A_i = 0 \rightarrow \sum_{i=1}^n (-1)^i \text{rank } A_i = 0,$$

as desired.

Now assume $H(A), H(X)$ are zero for $q > n$. For $q > n$, we have the exact sequence

$$0 = H_{q+1}(X) \rightarrow H_{q+1}(X, A) \rightarrow H_q(A) = 0,$$

so $H_q(X, A) = 0$ for $q > n + 1$. By the previous result on the long exact sequence $\cdots \rightarrow H_q(A) \rightarrow H_q(X) \rightarrow H_q(X, A) \rightarrow H_{q-1}(A) \rightarrow \cdots$, we have that $\chi(A) - \chi(X) + \chi(X, A) = 0$, ie $\chi(X, A) = \chi(X) - \chi(A)$, as desired.

2 Cell Complexes and more Adjunction Spaces

$S^2 = c^0 \cup c^2$. $Z_0 = c^0$. $f^0 = c^0 \cap \emptyset = \emptyset$. $\phi^0 : E^0 \rightarrow c^0$ is homeomorphism. $Z_1 = c^0$. $Z_2 = c^0 \cup c^2$. $f^2 = c^2 \cap c^0 = c^0$. $\phi^2 : E^2 \rightarrow S^2$ such that $E^2 - S^1 \xrightarrow{f^2} e^2 - e^0$ is homeomorphism.

3 Exact Sequences

Claim: Let $A \xrightarrow{f} B \xrightarrow{g} C$ be an exact sequence of R -modules, R an integral domain. Then $Q \otimes_R A \rightarrow Q \otimes_R B \rightarrow Q \otimes_R C$ is exact, where Q is the field of fractions of R .

Proof: Define $\varphi : Q \times A \rightarrow Q \otimes_R B$ by $\varphi(q, a) = q \otimes f(a)$. Note that $\varphi(qr, a) = qr \otimes f(a) = q \otimes rf(a) = q \otimes f(ra) = \varphi(q, ra)$, so φ factors through $Q \times A \rightarrow Q \otimes_R A$ to $f' : Q \otimes_R A \rightarrow Q \otimes_R B$, where $f'(q \otimes a) = q \otimes f(a)$ for all $q \in Q, a \in A$. The map $g' : Q \otimes_R B \rightarrow Q \otimes_R C$ is defined similarly.

Note that $g'f'(q \otimes a) = g'(q \otimes f(a)) = q \otimes gf(a) = q \otimes 0 = 0$. Suppose $g'b' = 0$ for some $b' \in Q \otimes_R B$. Write

$$\begin{aligned} b' &= \sum_{i \in [n]} q_i \otimes b_i = \sum_{i \in [n]} (m_i/n_i) \otimes b_i = \sum_{i \in [n]} (M_i/N) \otimes b_i \\ &= \sum_{i \in [n]} \frac{1}{N} \otimes (M_i b_i) = \frac{1}{N} \otimes \sum_{i \in [n]} M_i b_i. \end{aligned}$$

Hence, b' is of the form $q \otimes b$. Then $q \otimes g(b) = 0$.

When is the map $M \rightarrow Q \otimes_R M$ given by $m \mapsto 1 \otimes m$ injective? If $M \neq 0$ is a finite abelian group, $R = \mathbb{Z}, Q = \mathbb{Q}$, then $\mathbb{Q} \otimes_{\mathbb{Z}} M \cong 0$, so $M \rightarrow 0$ is not injective.

Let R be a commutative ring with nonzero elements $U = R - 0$. There is an isomorphism $Q \otimes_R M \rightarrow U^{-1}M$ given by $(r/s) \otimes m \mapsto (s, rm)$. Suppose $q \otimes g(b) = 0$. Let $q = r/s$ with $s \neq 0$. Then $(s, rg(b)) \sim (1, 0)$, so there exists $u \in U$ such that $us0 = u1rg(b) \rightarrow urg(b) = 0 = g(urb)$. By exactness of g ,

there exists $a \in A$ such that $f(a) = urb$, so

$$f'(q/(ur) \otimes a) = q/(ur) \otimes urb = q \otimes b.$$

4 Cell Complexes and More Adjunction Spaces

Claim: Let X, Y be Hausdorff. Then so is $X \vee Y$.

Proof: Let x_0 be the common point of $X \vee Y$. I claim that $X - x_0$ is open in $X \vee Y$. Indeed, let $p : X \sqcup Y \rightarrow X \vee Y$ be the quotient map. Then $p^{-1}(X - x_0) = X - x_0$ is open in $X \sqcup Y$, so $X - x_0$ is open in $X \vee Y$.

Let $x_1, x_2 \in X - x_0$ be distinct. Pick disjoint $U(x_1), V(x_2)$ in X , so $U' = U \cap (X - x_0), V' = V \cap (X - x_0)$ are disjoint open neighborhoods of x_1, x_2 in $X - x_0$. Then $U'(x_1), V'(x_2)$ are disjoint open in $X \vee Y$.

Let $x \in X - x_0$ and $y \in Y - x_0$. Then $U = X - x_0, V = Y - x_0$ works.

Finally, suppose $x = x_0$ and $y \in Y - x_0$ (the case where the other point is in $X - x_0$ is symmetric). Pick disjoint open $U(x_0), V(y)$ in Y . V is open in $Y - x_0$ and therefore $X \vee Y$. Let $U' = U \cup X$. I claim that U' is open in $X \vee Y$. Indeed, $p^{-1}U' = X \cup U$ is open in $X \sqcup Y$ and therefore $X \vee Y$. Further, since $(X \cup U) \cap V = \emptyset$ in $X \sqcup Y$ and p is surjective, we have that U', V are disjoint in $X \vee Y$.

Now assume x_0 has a neighborhood V that contracts to x_0 in Y . Consider the inclusion $(X \cup V, V) \rightarrow (X \vee Y, Y)$. Note that $\overline{Y - V} = Y - V$ in Y and Y is closed in $X \vee Y$, so $Y - V$ is closed in $X \vee Y$. Further, $Y^\circ = Y - x_0, Y$, which contains $Y - V$. Hence, $(X \cup V, V) \rightarrow (X \vee Y, Y)$ is an excision.

Since V contracts onto x_0 , there exists a homotopy equivalence $(X \cup V, V) \rightarrow (X, x_0)$ by fixing X . It follows that the inclusion $(X, x_0) \rightarrow (X \vee Y, Y)$ induces an isomorphism in homology, so the triple $(X \vee Y, X, Y)$ is exact.

By Mayer-Vietoris, we have the exact sequence

$$H_q^\#(x_0) \rightarrow H_q^\#(X) \oplus H_q^\#(Y) \rightarrow H_q^\#(X \vee Y) \rightarrow H_{q-1}^\#(x_0).$$

For $q \geq 0$, we have $H_q^\#(x_0) = 0$, so $H_q^\#(X \vee Y) = H_q^\#(X) \oplus H_q^\#(Y)$.

Let X, Y be Hausdorff. Let $X' = (X \vee Y)/Y$. Then $X \cong X'$.

Let $r : X \vee Y \rightarrow X$ be the retract given by $r(z) = z$ if $z \in X$ and $r(z) = x_0$ if $z \in Y$. Note that $X, Y \leq X \vee Y$ are closed since $p^{-1}(X) = X \sqcup x_0$ is closed in $X \sqcup Y$ (and similarly for Y). Let $q : X \vee Y \rightarrow (X \vee Y)/Y$ be the quotient map. Note that r sends Y to x_0 and is injective otherwise. Hence, r factors

through q to the map $f : (X \vee Y)/Y \rightarrow X$. Conversely, we have the maps $X \rightarrow X \vee Y \rightarrow (X \vee Y)/Y$, whose composition we denote by g . The maps f, g are inverses of each other, so $(X \vee Y)/Y \cong X$.

Let A, B, C be modules over a commutative ring R . Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be exact with A, C finitely generated. Then B is finitely generated.

Proof: Let $\{x_i\}_{i \in [n]}, \{z_j\}_{j \in [m]}$ generate A, C , respectively. Treat $A \leq B$, so $\varphi : B/A \cong C$. For all $j \in [m]$, pick a coset representative $y_j \in B$ of an element of $\varphi^{-1}(z_j)$. I claim that $\{x_i\}_{i \in [n]} \cup \{y_j\}_{j \in [m]}$ is a set of generators for B . Indeed, let $y \in B$. Write $\varphi(\bar{y}) = \sum_{j \in [m]} r_j z_j = \sum_{j \in [m]} r_j \varphi(\bar{y}_j) = \varphi\left(\overline{\sum_{j \in [m]} r_j y_j}\right)$, so $y - \sum_{j \in [m]} r_j y_j \in A$, so $y = \sum_{i \in [n]} s_i x_i + \sum_{j \in [m]} r_j y_j$, as desired.

Claim: Let R be a commutative ring. Let $\varphi : M \cong R$ as modules. Then there is a bijection Ψ between the set of submodules $N \leq M$ and the ideals $I \leq R$, given by $\Psi(N) = \varphi(N)$.

Let $N \leq M$ be a submodule. I claim that $I = \varphi(N) \trianglelefteq R$. Since $N \neq \emptyset$, $I \neq \emptyset$. Suppose $r, s \in I$. Then $r = \varphi(x), s = \varphi(y)$ for some $x, y \in M$. Then $r - s = \varphi(x - y) \in I$. Let $t \in R$ be arbitrary. Then $ts = t\varphi(x) = \varphi(tx) \in I$, so $I \trianglelefteq R$. Hence, Ψ is well-defined. Note if $\Psi(N) = \Psi(N')$, then $\varphi(N) = \varphi(N')$, so $N = N'$ since φ is a bijection. Hence, Ψ is an injection.

Finally, let $I \trianglelefteq R$. I claim that $N = \varphi^{-1}(I) \leq M$. Indeed, $N \neq \emptyset$. If $x, y \in N$, then $f(x), f(y) \in I$, so $f(x - y) = f(x) - f(y) \in I$, so $x - y \in N$. Finally, if $r \in R$, then $f(rx) = rf(x) \in I$, so $rx \in N$. Hence, $N \leq M$.

Claim: If R is Noetherian and Z is a spherical complex, then $H(Z)$ is finitely generated.

Proof: Let Y be a spherical complex. If Y has the finite discrete topology, then $H(Y)$ is finitely generated. Let Z be a spherical complex given by $Z = E^n \cup_f Y$ and assume $H^\#(Y)$ is finitely generated. We have $H_q^\#(Z) = H_q^\#(Y)$ for $q \neq n, n-1$, $H_{n-1}^\#(Z) \cong H_{n-1}^\#(Y)/\text{im } H_{n-1}^\#(f)$ and the exact sequence $0 \rightarrow H_n^\#(Y) \rightarrow H_n^\#(Z) \rightarrow \ker H_{n-1}^\#(f) \rightarrow 0$. The quotient of a finitely generated module is finitely generated, so $H_{n-1}^\#(Z)$ is finitely generated.

$\ker H_{n-1}^\#(f) \leq H_{n-1}(S^{n-1}) = R$, so by the previous lemma $\ker H_{n-1}^\#(f)$ corresponds to an ideal of R . Since R is Noetherian, I is finitely generated, so $\ker H_{n-1}^\#(f)$ is finitely generated. By the previous claim, $H_n^\#(Z)$ is finitely generated.

Let X be attached to Y at A by $f : A \rightarrow Y$. Suppose $g : X \rightarrow Z$ and $h : Y \rightarrow Z$ such that $g(a) = hf(a)$ for all $a \in A$. Then there exists a unique

continuous $k : X \cup_f Y \rightarrow Z$ such that the following diagram commutes:

$$\begin{array}{ccc} X \sqcup Y & \xrightarrow{g \sqcup h} & Z \\ & \searrow p & \uparrow k \\ & & X \cup_f Y \end{array}$$

Proof: since $g(a) = hf(a)$, $g \sqcup h$ factors through p to a map $k : X \cup_f Y \rightarrow Z$ such that the diagram commutes. Suppose $k_1, k_2 : X \cup_f Y \rightarrow Z$ satisfy $k_1 p = k_2 p = g \sqcup h$. Since p is surjective, we have $k_1 = k_2$, so k is unique.

Let X be Hausdorff. I claim that $(X \times I, X \times 0)$ is a collared pair. Indeed, $X \times I$ is Hausdorff, $A = X \times 0$ is closed in $X \times I$, the open neighborhood $B = X \times [0, 1/2)$ retracts onto A by $H : B \times I \rightarrow B$, $H((x, y), t) = (x, ty)$. If $(x, y) \notin A$, then $y \neq 0$. The neighborhoods $X \times [0, y/2)$, $X \times (y/2, 1]$ separate $A, (x, y)$.

The mapping cone CX is the quotient of CX by $X \times 0$. In particular, take the constant adjunction map $f : X \times 0 \rightarrow z_0$, so (CX, z_0) is a collared pair, so CX is Hausdorff.

Let A be a strong deformation retract of the set B . Denote by $\{A_\alpha\}_{\alpha \in J}, \{B_\beta\}_{\beta \in K}$ the components of A, B . Let $H : B \times I \rightarrow B$ be the SDR.

$H(B_\beta \times 1) \leq A$ is connected, so $H(B_\beta \times I) \leq A_\alpha$ for a unique $\alpha \in J$. Denote by $\varphi : K \rightarrow J$ this assignment. Conversely, $A_\alpha \leq B$ is connected, so there exists a unique $\beta \in K$ such that $A_\alpha \leq B_\beta$. Denote by $\psi : J \rightarrow K$ this assignment. I claim that φ, ψ are inverses.

We have $H(B_\beta \times 1) \leq A_{\varphi(\beta)} \leq B_{\psi\varphi(\beta)}$. Since $B_\beta \times I$ is connected and $H(B_\beta \times I) \cap B_{\psi\varphi(\beta)} \neq \emptyset$, we have $H(B_\beta \times I) \leq B_{\psi\varphi(\beta)}$. In particular, $B_\beta = H(B_\beta \times 0) \leq B_{\psi\varphi(\beta)}$, so $\psi\varphi = 1$.

Let $\alpha \in J$. Then $A_\alpha \leq B_{\psi(\alpha)}$. We have $H(B_{\psi(\alpha)} \times 1) \geq A_\alpha$, but $H(B_{\psi(\alpha)} \times 1)$ is connected, so $H(B_{\psi(\alpha)} \times 1) = A_\alpha$. It follows that $\varphi\psi(\alpha) = \alpha$, so ψ, φ are inverses.

Consider the restriction $H_\beta : B_\beta \times I \rightarrow B$. Note that $H_\beta(B_\beta \times I) \leq B_\beta$, so we can restrict the image to B_β . We have $A_{\varphi(\beta)} \leq B_\beta$ and $H_\beta(B_\beta \times 1) \leq A_{\varphi(\beta)}$, so $A_{\varphi(\beta)}$ is a SDR of B_β . Since φ, ψ are inverses, there exists some component B_β that SDR's onto A_α for every $\alpha \in J$.

Consider the space $X \times I$. Let $A = (X \times 0) \cup (X \times 1)$, so (X, A) is collared. Then in CX , we have $(CX, x_0 \cup (X \times 1))$ is collared. I might as well show explicitly (CX, X) is collared.

Given $f : X \rightarrow Y$, we obtain $Cf : CX \rightarrow CY$ (C is functorial?)

Proof: Define $f \times 1 : X \times I \rightarrow Y \times I$. Note that the fiber $X \times 0$ is sent to $Y \times 0$. Hence, f factors through the quotient maps to $Cf : CX \rightarrow CY$.

Claim: The vertex is a SDR of CX .

Proof: $A = X \times 0$ is a SDR of $X \times I$. Let $H : (X \times I) \times I \rightarrow X \times I$ be this SDR. Let $p : X \times I \rightarrow CX$ be the quotient map. We have the induced map $\hat{H} : CX \times I \rightarrow CX$ that is the SDR of CX onto the vertex x_0 .

Claim: $f : X \rightarrow Y$ is nullhomotopic iff f extends to $F : CX \rightarrow Y$.

Proof: Suppose f extends to $F : CX \rightarrow Y$. Let $p : X \times I \rightarrow CX$ be the quotient map. Define $H : X \times I \rightarrow Y$ by $H(x, t) = Fp(x, t)$. Then $Fp(x, 0) \in Y$ is constant and $Fp(x, 1) = f(x)$ by assumption, ie f is null-homotopic.

Conversely, if $f : X \rightarrow Y$ is null-homotopic by the homotopy $H : X \times I \rightarrow Y$, then H is constant on the fibers of p , ie $X \times 0$, and gives a map $f : CX \rightarrow Y$ extending f .

Define $\gamma : Cf \rightarrow Cf'$ by

$$\begin{aligned} (x, t) &\mapsto \begin{cases} (\alpha(x), 2t) & \text{if } 0 \leq t \leq \frac{1}{2}, \\ e'H(x, 2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1, \end{cases} \\ e(y) &\mapsto e'\beta(y). \end{aligned}$$

The result follows when we show $\gamma\bar{f} \simeq \bar{f}'\hat{\alpha}$. The required homotopy $K : (CX \times I, X \times I) \rightarrow (Cf', Y')$ is given by

$$K(x, t, s) = \begin{cases} (\alpha(x), 2t/(s+1)) & \text{if } 0 \leq t \leq \frac{s+1}{2}, \\ e'H(x, 2t - s - 1) & \text{if } \frac{s+1}{2} \leq t \leq 1. \end{cases}$$

Consider the space X of two interlocking tori tangent to each other at $S^1 \vee S^1$. Consider the inclusion $f : S^1 \vee S^1 \rightarrow S^1 \times S^1$. $S^1 \times S^1 - S^1 \vee S^1 \cong (0, 1)^2$ is mapped homeomorphically to $X - S^1 \times S^1$. Hence, we can identify $X \cong (S^1 \times S^1) \cup_f (S^1 \times S^1)$.

Let $X = E^2 \times S^1$, so $\partial X = S^1 \times S^1$. Let $\lambda : S^1 \times S^1 \rightarrow S^1 \times S^1$ be given by $\lambda(e^{i\theta}, e^{i\phi}) = (e^{i(\theta+\phi)}, e^{i\phi})$ and $\mu(e^{i\theta}, e^{i\phi}) = (e^{i\theta}, e^{i(\theta+\phi)})$. Let $M = S^1 \times x_0$ and $L = x_0 \times S^1$. Note that $\lambda|M = 1$ and $\mu|L = 1$.

Identify S^3 by $(z_0, z_1) \in \mathbb{C}^2$ such that $|z_0|^2 + |z_1|^2 = 1$. Let $p, q \in \mathbb{Z}^+$ be coprime and let $h : S^3 \rightarrow S^3$ be given by $h(z_0, z_1) = (z_0 e^{2i\pi/p}, z_1 e^{2i\pi q/p})$ be a homeomorphism of order p . h acts properly discontinuously on S^3 . Let $L(p, q)$

denote the resulting quotient space by $\langle h \rangle$.

If $p = 1$, then $h(z_0, z_1) = (z_0, z_1)$ is the identity, so the quotient space on S^3 is itself.

$$H_2(E^2 \times S^1) \oplus H_2(E^2 \times S^1) \rightarrow H_2(L) \rightarrow H_1(S^1 \times S^1) \rightarrow H_1(E^2 \times S^1) \oplus H_1(E^2 \times S^1) \\ 0 \rightarrow H_2(L) \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$$

The rightmost map has the matrix

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

so $H_2(L) = \mathbb{Z}$?

5 Topology of $\mathbb{R}$

Does every closed subset A of $\mathbb{R}$ have an open neighborhood U that SDR's onto A ?

Answer: Consider $A = \{0\} \cup K$, where $K = \{1/n : n \in \mathbb{Z}^+\}$. Let $U(A)$ be open, $H : U \times I \rightarrow U$ be the SDR. Each element of A is in its own component. By the previous lemma, U has components $\{U_n\}_{n \in \mathbb{Z}_{\geq 0}}$, where U_0 SDR's onto $\{0\}$ and U_n SDR's onto $\{1/n\}$ for $n > 0$. Since $\mathbb{R}$ locally connected and U is open, each component U_i is open. In particular, U_0 is open. Since $0 \in \bar{K}$, there exists $N \in \mathbb{Z}^+$ such that $1/N \in U_0$, absurd. Note that A is also compact.

Suppose $A \leq \mathbb{R}^n$ is bounded. Then $\bar{A} \leq \mathbb{R}^n$ is bounded.

Proof: Write $A \leq [-M, M]^n$ for some $M \in \mathbb{R}^+$. Then $\bar{A} \leq [-M, M]^n$, so $\bar{A}$ is bounded.

Claim: Let $f : \mathbb{R}^n \hookrightarrow S^n$. $A \leq \mathbb{R}^n$ is bounded iff $\infty \notin \overline{f(A)}$ in S^n .

Proof: Suppose $A \leq S^n$ is bounded. There exists $\epsilon > 0$ such that $A \leq D_\epsilon(0)$, so in S^n we have $\infty \in S^n - f(D_\epsilon(0)) = U(\infty)$. Note that U is open since $D_\epsilon(0)$ is compact.

If $A \leq S^n$ is unbounded, pick $x_n \in A$ such that $|x_n| \geq n$. Then $f(x_n) \rightarrow \infty$, so $\infty \in \overline{f(A)}$.

Claim: Let $U \leq \mathbb{R}^n$ be open and $x_0 \in U$. Then $x_0 \in \overline{U - x_0}$.

Proof: Pick $B(x_0, \epsilon) \leq U$. Then $x_0 \in \overline{B(x_0, \epsilon)} - x_0 = \overline{B(x_0, \epsilon)} \leq \overline{U}$.

Let $f : S^{n-1} \rightarrow \mathbb{R}^n$ be an embedding. Denote by s_{n-1} the image. Include $\mathbb{R}^n \rightarrow S^n$ and let $S^n - s_{n-1} = C_0 \cup C_\infty$, so C_0, C_∞ are open. Since C_∞ is open and $\infty \in C_\infty$, we have $\infty \in \overline{C_\infty} - \infty$. It follows that $C_\infty - \infty$ is unbounded. Also, $\infty \notin \overline{C_0} - \infty = S^n - C_\infty$, so C_0 is bounded.

Claim: Let $n > m$, $U \leq \mathbb{R}^n$ be open. There exists no injection $f : U \hookrightarrow \mathbb{R}^m$.

Proof: Consider the composition $U \xrightarrow{f} \mathbb{R}^m \hookrightarrow \mathbb{R}^n$. It is assumed $U \neq \emptyset$, so pick $x_0 \in U$ and $E^n(x_0) \leq U$. Thus, we have the injection $E^n \hookrightarrow \mathbb{R}^n$. By Invariance of Domain, $(E^n)^\circ$ is sent to an open set of $\mathbb{R}^n$. However, under the composition, $(E^n)^\circ \mapsto A$ for some $A \leq \mathbb{R}^m$, absurd.

Claim: Let X be a topological manifold and $x_0 \in X$. There exists a unique integer n such that x_0 has a neighborhood homeomorphic to $\mathbb{R}^n$.

Proof: Suppose FTSOC x_0 has neighborhoods U, V such that $h : U \rightarrow \mathbb{R}^n, k : V \rightarrow \mathbb{R}^m$ are homeomorphisms. Let $W = h^{-1}(U \cap V)$. We have the composition

$$W \xrightarrow{\sim} U \cap V \hookrightarrow V \xrightarrow{\sim} \mathbb{R}^m,$$

ie an injection $W \hookrightarrow \mathbb{R}^m$, absurd. Denote by $\dim x$ this unique $n \in \mathbb{Z}^+$.

For each $n \in \mathbb{Z}^+$, let $X_n = \{x \in X : \dim x = n\}$. I claim that $X_n \leq X$ is open. Suppose $x \in X_n$. Pick $U(x)$ such that $h : U \rightarrow \mathbb{R}^n$ is a homeomorphism. Let $y \in U$. Then $B_1(hy) \cong \mathbb{R}^n$, so $h^{-1}B_1(hy)$ is a neighborhood of y isomorphic to $\mathbb{R}^n$, ie $y \in X_n$.

Note that $X = \bigcup_{n \in \mathbb{Z}^+} X_n$ is a partition of X into open subspaces. Since X is connected, there exists a unique $n \in \mathbb{Z}^+$ such that $X = X_n$. Let $n = \dim X$.

Claim: The boundary ∂X of a manifold with boundary X is a manifold.

Proof: $\partial X \leq X$ is Hausdorff second countable. Let $x \in \partial X$, so there exists $U(x) \cong \mathbb{R}^n \times \mathbb{R}_{\geq 0}$ for some $n \geq 0$. Note that if $h(y) \in \mathbb{R}^n \times \mathbb{R}^+$, then y has a neighborhood homeomorphic to $\mathbb{R}^{n+1}$. Hence, $U \cap \partial X \leq \mathbb{R}^n \times 0$. It is clear that any element of the preimage of $\mathbb{R}^n \times 0$ is in ∂X , so $U \cap \partial X \cong \mathbb{R}^n \times 0 \cong \mathbb{R}^n$, ie if $\dim X = n + 1$, then $\dim \partial X = n$.